Scale-Invariant Safety Geometry: Universal Vulnerability Across LLM Architectures

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Abstract

We demonstrate that safety mechanisms in large language models share a **universal geometric structure** that becomes apparent only after proper normalization. Through systematic experiments across four model families (Llama-3.1, Gemma-2, Qwen-2.5, Mistral), we show that apparent differences in steering resistance are explained entirely by the first singular value (σ_1) of the safety subspace—not by qualitative architectural differences. When perturbation magnitude is normalized by σ_1, all tested models exhibit identical vulnerability profiles: 0% bypass at $\alpha_{\text{eff}} = 0$, 100% bypass at $\alpha_{\text{eff}} \geq 5$. This **scale-invariant safety geometry** has profound implications: (1) models previously thought "robust" are simply operating at different scales; (2) safety training creates similar geometric structures regardless of architecture; (3) red-teaming methodologies must account for model-specific scaling factors. We introduce $\alpha_{\text{eff}} = \alpha/\sigma_1$ as the universal metric for comparing steering attacks across models.

Keywords: Activation steering, AI safety, Singular value decomposition, Universal vulnerability, Mechanistic interpretability

1 Introduction

Recent work on activation steering has shown that language model behaviors can be manipulated by adding vectors to intermediate activations. A natural question is whether safety mechanisms are equally vulnerable. We resolve apparent inconsistencies across prior work with a simple but powerful finding: **all tested models are equally vulnerable when perturbations are normalized by the safety subspace's spectral norm**.

Consider two models:

- **Llama-3.1-8B**: $\sigma_1 \approx 5$, bypassed at $\alpha_{\text{raw}} = 25$
- **Gemma-2-9B**: $\sigma_1 \approx 100$, bypassed at $\alpha_{\text{raw}} = 500$

The ratio of thresholds (20x) matches the ratio of σ_1 values. When we define **effective steering magnitude**:

$$\alpha_{\text{eff}} = \frac{\alpha_{\text{raw}}}{\sigma_1} \tag{1}$$

both models bypass at $\alpha_{\text{eff}} \approx 5$.

Model/Layer	σ_1	**Eff. Rank**
Mistral L16	2.8	16
Llama L16	4.5	16
Llama L24	9.8	16
Qwen L14	14.9	16
Qwen L21	47.9	16
Gemma L21	101.6	16
Gemma L31	178.0	16

Table 1: σ_1 varies by 60x while effective rank remains constant.

Model	**Bypass Rate by α_{eff}**						
	0	5	10	15	20	25	30
Llama	0%	100%	100%	100%	100%	100%	100%
Gemma	0%	100%	100%	100%	100%	100%	100%
Qwen	0%	100%	100%	100%	100%	100%	100%
Mistral	38%	100%	100%	100%	100%	100%	100%

Table 2: When normalized by σ_1, all models show identical vulnerability.

2 Results

2.1 Cross-Model Geometry

2.2 Universal Vulnerability

3 Conclusion

Safety mechanisms share universal geometric structure. Apparent robustness differences are explained entirely by σ_1. We introduce $\alpha_{\text{eff}} = \alpha/\sigma_1$ as the universal metric for cross-model comparison.